

Primality

AA-GEI FIB, UPC

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The Prime Number Theorem

Gauss, 1793 conjectured the result:

Let $n \in \mathbb{Z}^+$ and let $\pi(n)$ be the number of primes $\leq n$, then

$$\left[\frac{\text{number-prime less } n}{\frac{n}{\ln n}} = \text{fairly close!} \right] \lim_{n \rightarrow \infty} \frac{\pi(n)}{n / \ln n} \rightarrow 1.$$

Let p_n be the n -th. prime number, then the value of $p_n \sim n \ln n$

Fermat's Little Theorem and Luca's characterization

Proposition (Fermat's Little Theorem (FLT) (2.3.11))

For any n prime and for all $a \in \mathbb{Z}_n^+$, $a^{n-1} \equiv 1 \pmod{n}$.

Proposition (Fermat's Little Theorem (FLT) (2.3.12))

Let $a, n \in \mathbb{Z}^+$, then n is prime iff:

- ❶ $a^{n-1} \equiv 1 \pmod{n}$, and
- ❷ for all prime divisors q of $n - 1$, we have $a^{(n-1)/q} \not\equiv 1 \pmod{n}$.

Primality problems

- **Primality problem:** given $n \in \mathbb{Z}^+$ with N -bits, is n prime?
The Primality problem belongs to P
- Given an integer n , can we generate in polynomial time the n -th prime number?
- Given an integer N , can we generate in polynomial time a N -bit prime number?

The Primality problem: a naive algorithm

Naive algorithm:

```

Is  $n \in \mathbb{N}$  prime?
for  $a = 2, 3, \dots, \sqrt{n}$  do
    if  $a \mid n$  then
        return "composite", and
        STOP
return prime and STOP
  
```

Notice: for large n ($n = 2^{2024}$)
 the input is $N = \lg n$ i.e.
 $n = 2^N$

May need to check $O(\sqrt{n})$
 integers for divisors of n , each
 check with a cost $O(N^2)$
 $T(N) = O(2^{N/2} N^2)$ Too slow!

Test of pseudo-primality

n is a **base a pseudo-prime** if $a^{n-1} \equiv 1 \pmod{n}$.

PseudoPrime(n)

$a = \text{random}(1, n - 1)$

if $a^{n-1} \equiv 1 \pmod{n}$ **then**

return pseudo-prime

else

return not-prime

Cost: $O(N^3)$.

Test of pseudo-primality

n is a **base a pseudo-prime** if n is composite and $a^{n-1} \equiv 1 \pmod{n}$.

PseudoPrime(n)

$a := \text{random}(1, n-1)$

if $a^{n-1} \equiv 1 \pmod{n}$ **then**

return pseudo-prime

else

return not-prime

If the algorithm says not-prime, it is correct. It can err only if n is base 2 pseudo prime. Can we bound the error?

Carmichel numbers

- Recall Fermat's theorem: If n is prime, for $a \in \mathbb{Z}_n^+$, $a^{n-1} \equiv 1 \pmod n$
- BUT** $\exists n \in \mathbb{Z}$ s.t., for $a \in \mathbb{Z}_n^+$, $a^{n-1} \equiv 1 \pmod n$ with n **NOT** a prime:
Carmichael number. *Complex composites that behave as if they were prime!*
- Carmichael numbers are very rare (255 with value $< 10^8$) 561, 1105, 1729, $\dots$. For example $561 = 3 \cdot 11 \cdot 17$.
- A numerical conjecture is that the number of Carmichael numbers less than n is $\leq n^{2/7}$. (OEIS)

*growing much
faster than
with parameter*

Lowering the error probability

- If we assume the **non existence of Carmichael numbers**: if n is prime, **PseudoPrime** always gives the correct answer, but if n is composite it errs in telling so with probability $\leq 1/2$.
- **PseudoPrime** has one-side error, therefore *amplifying* k times the algorithm, the probability of error goes down to $\leq 1/2^k$.

Taking into account the Carmichael numbers

- $x \in \mathbb{Z}$ is a **non-trivial root of 1 mod n** , if it satisfies $x^2 \equiv 1 \pmod{n}$, but $x \not\equiv \pm 1 \pmod{n}$.
- **Example** 6 is a non-trivial root of 1 mod 35: as $6^2 \equiv 1 \pmod{35}$ but $6 \not\equiv \pm 1 \pmod{35}$.

Taking into account the Carmichael numbers

Theorem

If $p > 2$ is prime and $n = p^k$, for $k \geq 1$, the equation $x^2 \equiv 1 \pmod n$ has only two solutions $x = 1$ and $x = -1$.

Corollary

If there exists a non-trivial square root of 1 mod n , then n is composite.

Miller-Rabin: Randomized Algorithm

G. Miller (1976), M. Rabin (1980)

- If equation $x^2 \equiv 1 \pmod{n}$ has exactly solutions $x = \pm 1$ that implies n is prime.
- If there is another solution different than ± 1 , then n can not be prime.
- To see if n is prime:
Randomly choose an integer $a < n$, if $a^2 \equiv 1 \pmod{n}$, then a is a non-trivial root of $1 \pmod{n}$, so n is not prime. Such an n is denoted a **witness** to the compositeness of n .
Otherwise, n may be or may be not a prime.

Miller-Rabin: Randomized Algorithm

- generate a random value $a \in \mathbb{Z}_n^+$,
- check if $a^{n-1} \equiv 1 \pmod{n}$,
- in doing so look if an integer x s.t. x is a non-trivial square root of 1 mod n can be found.
- Repeat to decrease the probability of error.

Witness to the compositeness of n

Given an odd integer $n > 2$, and $a \in \mathbb{Z}_n^+$, we say that a is a witness to the compositeness of n , if either:

- $a^{n-1} \not\equiv 1 \pmod{n}$
- $\exists m \in \mathbb{Z}_n^+$ s.t. $x = a^m$ is a non-trivial square root of $1 \pmod{n}$

We define a function **Witness** (a, n) to test if $a^{n-1} \not\equiv 1 \pmod{n}$ or if we can find a non-trivial root of $1 \pmod{n}$.

Witness (a, n)

Witness(a, n)

compute t and u s.t. $n - 1 = 2^t u$

$x_0 = a^u \bmod n$

for $i = 1$ to t **do**

$x_i = x_{i-1}^2 \bmod n$

if $x_i = 1 \wedge x_{i-1} \neq 1 \wedge x_{i-1} \neq n - 1$ **then**

return true { x_i is a non-trivial root of 1}

if $x_t \neq 1$ **then**

return true {Fermat's fails}

return false { a is not a witness}

Example: Wish to test if $a = 7$ is a witness to $n = 561$

$$N - 1 = 560 = \underbrace{100011}_{u} \underbrace{0000}_{2^t} \Rightarrow u = 35, t = 4$$

$$x_0 = 7^{35} \bmod 561 = 241$$

$$x_1 = 241^2 \bmod 561 = 298$$

$$x_2 = 298^2 \bmod 561 = 166$$

$$x_3 = 166^2 \bmod 561 = 67$$

$$x_4 = 67^2 \bmod 561 = 1$$

Non-trivial root of 1 mod 561.

The cost of **witness**(a, n) is $O(N^3)$.

Miller-Rabin primality test.

Polynomial time randomized algorithm to decide if a given $n \in \mathbb{Z}$ is prime. The input to the algorithm would be n and the number s of $a \in \mathbb{Z}$ that we will test for witness.

```

Miller-Rabin( $n, s$ )
  for  $i = 1$  to  $s$  do
     $a = \text{random}(1, n - 1)$ 
    if witness ( $a, n$ ) = true then
      return non-prime {Definitely}
  return prime. {Almost surely}
  
```

The complexity of the algorithm is $O(sN^3)$.

Correctness

If n is an odd composite number, the number of witnesses to the compositeness of n is $\geq \frac{n-1}{2}$.

For any odd integer $n > 2$ and $s \in \mathbb{Z}^+$ the probability that Miller-Rabin(n, s) errs is $\leq 2^{-s}$.

- If n is composite, Miller-Rabin errs if it misses to discover a witness in the s iterations.
- If n is composite, each execution of the algorithm has probability $\geq 1/2$ of discovering a witness a .
- The probability it misses in all iterations is $< 1/2^{-s}$.

Procedure to Generate a large Prime Integer n

- ① Choose a random odd N -bit number n ,
- ② Run Miller-Rabin on n , if passes, output n ,
- ③ else repeat the process at most s times.

With probability $O(1/N)$, n will be prime $\Rightarrow n$ will pass Miller-Rabin.

Using Bayes, the probability that n is prime given that Miller-Rabin has returned prime is $\frac{1}{1+2^{-s}(\ln n - 1)}$

Which is below $1/2$ until s exceeds $\log(\ln n - 1)$.

In practice, $s = 50$ suffices for most of the practical applications.